Rove Generation

AURAL ENHANCER

Aural Exciter & Optical Big Bottom

User Manual

Overview

Aural Enhancer is a dual-process audio enhancement plugin inspired by the classic Aphex hardware units (Model 204, Type C2, and Type III). It pairs two independent processors - an Exciter for the highs and a Big Bottom for the lows - each running as a parallel sidechain alongside the untouched dry signal.

The Exciter side works by generating new harmonic content above the source material's existing spectrum. Rather than simply turning up what's already there (as an EQ would), it synthesizes harmonics that weren't in the original recording. The result is more sparkle, definition, and "air" - without pushing the output level up in any meaningful way.

The Big Bottom side takes a completely different approach. Instead of boosting bass amplitude, it manipulates the timing and dynamics of low-frequency content to create the perception of a heavier, more sustained low end. Your speakers and amplifiers don't work any harder, but the bass feels significantly bigger.

Both processors can be used independently or together. Additional features include a Spectral Phase Refractor (SPR) for bass phase correction on the main audio path, a mastering-grade 6 dB/oct input high-pass filter, a Timbre control for shaping the harmonic character, a Density mode for denser harmonic generation, continuous Peaking and Null Fill controls for precise filter shaping, Phase and AutoTrace switches for the Big Bottom section, Solo modes for monitoring each sidechain, up to 4x oversampling, a latency-compensated wet/dry mix, and a soft-clip output limiter.

Signal Flow

Input → HPF → SPR → Input Gain → [Exciter + Big Bottom + Dry] → Sum → Output Gain → Clip → Mix → Output
The plugin first passes your input through an optional 6 dB/oct high-pass filter (HPF) for subsonic cleanup, then through the optional Spectral Phase Refractor (SPR) which corrects bass phase alignment on the main audio path. The signal then splits into three parallel paths. The dry signal passes straight through untouched. The Exciter sidechain taps the input through a high-pass filter, generates harmonics, and feeds a small amount back. The Big Bottom sidechain taps through a low-pass filter, shapes the phase and dynamics, and feeds that back too. The final output is all three paths summed together. Because both sidechains run in parallel (not in series), they don't interfere with each other. Each one works on its own frequency territory independently.

Big Bottom

The Big Bottom takes an entirely different approach from the Exciter. Where the Exciter creates new harmonic content, the Big Bottom reshapes the perception of existing bass through timing and dynamics. It isolates the low end with a tunable low-pass filter (20-400 Hz), runs it through a short delay that tracks the filter frequency, and then through a leveling stage that keeps the output at a near-constant level regardless of how hard the input is hitting.

The clever part is the leveling stage. When your input is quiet, the processed bass represents a bigger share of the mixed output - so the bass fills in. When the input is loud, the processed signal becomes a tiny fraction of the total - so the peaks barely change. You get the sensation of deeper, more sustained bass without actually pushing your levels up. That means no extra clipping risk, no overdriven speakers, and no headroom problems.

The delay is what gives Big Bottom its characteristic “hang” on bass notes - the delayed copy extends the duration of the low end, making it feel weightier and more present in the mix.

Big Bottom Controls

Drive	Sets the intensity of the bass processing (0-100%). Higher settings mean more sustain and a denser bass effect. Lower settings are more transparent. Find the sweet spot where the bass notes feel longer and more solid without becoming muddy.
Tune	Low-pass filter frequency (20-400 Hz). At noon this sits around 90 Hz. Turn left to focus on deep sub-bass; turn right to bring the effect up into upper bass and low-mid territory. The right setting depends on your

	material – kick-heavy music might want 80-100 Hz, while bass guitar work might benefit from 120-150 Hz. The extended upper range up to 400 Hz lets you reach into upper bass and low-mid territory for toms, cello, and male vocals.
Amount	How much processed bass is mixed into the output (0 - 100%). Moderate settings give you the magic - perceptibly heavier bass without the meters moving. Push it too far and you will start to see peak levels climb, so keep it reasonable.
Process	On/off switch for the Big Bottom sidechain. When off, no bass enhancement is applied.
Peaking	Switches the low-pass filter between a gentle wide slope (off) and a focused resonant peak (on) at the Tune frequency. Narrows the bass enhancement to a tighter frequency band around your chosen point.
Solo	Mutes the dry signal and outputs only the Big Bottom sidechain. Useful for hearing exactly what the bass processing is adding. Turn off when done tuning. Requires Process to be on.
Phase	Inverts the polarity of the Big Bottom sidechain before it is mixed back with the dry signal. Off gives the classic Model 104 sound. On flips the phase relationship, which fundamentally changes the tonal character of the bass enhancement - some material responds better to one setting than the other. Note that Phase on may increase the output level, so you may need to reduce Amount or Drive to compensate.
AutoTrace	Enables self-optimizing dynamics in the leveling amplifier. When on, the processor tracks the input level and automatically adjusts its internal threshold so the bass enhancement stays consistent regardless of how loud or quiet the source material is. Drive becomes less sensitive and the effect works across a wider dynamic range. Particularly useful for full mixes, live recordings, or any material where the bass level varies over time.

Aural Exciter

The Exciter works as a high-frequency sidechain. Your signal is tapped through a tunable high-pass filter (600 Hz - 7 kHz), which selects the frequency range that will be “excited.” This filtered signal is then processed to generate new harmonics, paying special attention to transients - sharp attacks get more harmonic emphasis, while smooth sustained tones are treated more gently. This keeps things sounding natural rather than harsh.

The Timbre knob controls the balance between warm, smooth harmonics (even) and brassy, cutting harmonics (odd). An alternative gentler mode is available via the Clip switch, which produces purer, silkier harmonics. The Density switch offers a High mode that compresses peaks and expands quiet signals for denser, more consistent harmonic generation. The Peaking control varies the sidechain filter’s

resonance from a gentle slope to a focused peak at the Tune frequency, while Null Fill restores presence at the Tune frequency that can get lost when the sidechain is mixed back in. The generated content is blended into the original signal at a much lower level - you hear it as added clarity and dimension, not as distortion.

Exciter Controls

Tune	Corner frequency of the sidechain high-pass filter, from 600 Hz (fully left) to 7 kHz (fully right). At noon it sits around 4 kHz. Lower settings pull in more midrange for a “forward” presence effect. Higher settings target only the top end for an open, airy quality.
Harmonics	How aggressively new harmonic content is generated (0-100%). Low settings are subtle - good for vocals and full mixes where you want transparency. High settings add real bite and edge, great for drums, guitars, and synths that need to cut through.
Timbre	Controls the harmonic character (0-100%). At 0% (EVEN) you get both even and odd harmonics for a warm, smooth sound. At 100% (ODD) only odd harmonics are produced, giving a brassy, aggressive character. Settings in between give a proportional blend.
Amount	How much of the generated harmonics get blended into the output (0-100%). A little goes a long way. Start low, increase until you notice added clarity, and stop before it gets harsh.
Process	On/off switch for the Exciter sidechain. When off, no high-frequency enhancement is applied.
Peaking	Continuously varies the sidechain filter’s resonance (0-100%). At 0% the filter has a gentle slope - smooth and uncolored. Around 50% you get a balanced, flat response. At 100% there’s a strong resonant peak (+6 dB) right at the Tune frequency, focusing harmonic generation on a narrow band. Inspired by the Aphex Type III PEAKING control.
Clip	Selects the harmonic generation mode. Cubic is the default - it produces brassy, aggressive harmonics with more bite. Sin uses a gentler approach that produces only pure harmonics with a smoother, silkier character. Both respond to the Timbre control for even/odd harmonic balance.
Null Fill	Restores presence at the Tune frequency (0-100%). When the exciter’s sidechain is mixed back with the dry signal, a dip naturally occurs at the Tune frequency. Null Fill compensates for this by adding energy right at that frequency, restoring presence and body without sacrificing the high-frequency extension above. Inspired by the Aphex Type III NULL FILL control. At 0% you get a pure high-pass response; higher settings progressively fill the dip.

Solo	Mutes the dry signal and outputs only the Exciter sidechain. Useful for dialing in the Tune, Peaking, and Null Fill settings - you hear exactly what the exciter is generating. Turn off when done tuning. Requires Process to be on.
Density	Selects the harmonic generation density mode. Normal passes the signal through the waveshaper unmodified. High compresses peaks and expands low-level signals before the waveshaper, producing a higher density of harmonics with better peak control. Use High when you want a hotter, more consistent harmonic character, or when clipping occurs at high Harmonics settings.

Global Controls

HPF	Master input high-pass filter (0–200 Hz). A gentle 6 dB/oct slope removes subsonic rumble before it reaches the processors. At 0 Hz the filter is bypassed. Turn it up to set the cutoff frequency. Useful for cleaning up accumulated low-end buildup on mix buses and mastering chains, ensuring Big Bottom works on real bass content rather than subsonic noise.
SPR	Spectral Phase Refractor. Processes the main audio signal so that bass frequencies (up to approximately 150 Hz) lead in phase relative to the rest of the spectrum. This corrects the cumulative phase smearing that occurs through recording, processing, and reproduction, restoring clarity, openness, and apparent bass energy without adding any amplitude boost. Works independently of both sidechains – you can use SPR with both processors turned off. Most effective on heavily processed material, overdubbed tracks, live recordings, and reverberant sources. Inspired by the Aphex Type III Spectral Phase Refractor.
Input Level	Gain before processing (-20 to +20 dB). Pushing the input hotter drives both sidechains harder for a more pronounced effect. For subtle enhancement, stay near 0 dB.
Output Level	Gain after processing (-20 to +20 dB). Use this to level-match your processed signal against the bypassed signal, so you can A/B without being fooled by volume differences.
Mix	Wet/dry blend (0-100%). At 100% you hear only the processed output. Lower values blend in the original dry signal for parallel processing.
Quality	Oversampling mode. Normal (1x) is CPU-friendly and adds zero latency. HQ (2x) and Ultra (4x) run the processing at a higher internal sample rate, which reduces aliasing artifacts from the harmonic generation.
Clip 0dB	Catches any peaks that exceed 0 dBFS when you're driving the plugin hard. Keeps your output clean without needing a separate limiter.

Bypass	Smoothly fades between processed and dry over about 3 ms to avoid clicks. When bypassed, your signal passes through completely unchanged.

Quick Start

A good approach when dialing in is to work on each processor separately. Turn BB Amount to zero while you set up the Exciter, then turn Exc Amount to zero while you shape the Big Bottom. Once both sound right on their own, bring them both up and use Bypass to compare against the original.

Thicker bass: BB Drive 30-40%, BB Tune around 100 Hz, BB Amount 30-50%. Listen for the low notes becoming longer and fuller. Adjust Drive until you feel the bass “hang” a bit on each hit.

Vocal clarity: Exc Tune 2-4 kHz, Harmonics 20-40%, Amount 20-40%, Timbre 0-30%. Vocals gain definition and cut through the mix more easily, without actually being louder.

Air and sparkle: Exc Tune 5-7 kHz, Harmonics 30–50%, Amount 15-30%. Creates shimmer and openness in the top end, especially nice on acoustic instruments and overheads.

Aggressive edge: Push Timbre to 70-100% to lean into odd harmonics. Sounds brassy and forward useful on electric guitars, snares, and synth leads.

Smooth shimmer: Switch Exc Clip to Sin for a gentler harmonic character. The Sin mode produces only soft harmonics - great for vocals, acoustic instruments, and master bus sweetening where you want presence without any harshness.

Focused excitation: Turn up Peaking to 60-80% to concentrate harmonic generation around the Tune frequency. Adds a resonant emphasis that makes the exciter target a specific band more aggressively. Combine with Null Fill at 30-50% to restore presence in the dip below the peak.

Narrowband targeting: Enable Peaking on either section to tighten the sidechain filter. Use Solo to hear exactly what the sidechain is generating while you dial in Tune, Peaking, and Null Fill. Turn Solo off when done.

Parallel processing: Set Mix to 30-50% and push both processors harder than you normally would. The dry signal preserves your transients while the wet signal adds body and sheen. Works especially well on mix buses and drum groups.

Output protection: Switch on Clip 0dB whenever you’re driving the input or running both processors at high Amount settings. It catches overs gracefully.

Best quality: Set Quality to Ultra (4x) for final mixes. The extra oversampling cleans up the harmonic generation and gives you the most transparent result.

Subsonic cleanup: Set HPF to 30–40 Hz for gentle rumble removal on mix buses, or 60–80 Hz for tighter bass on non-bass tracks. This ensures Big Bottom processes real musical content rather than subsonic noise. Leave at 0 Hz when no filtering is needed.

Phase correction: Switch on SPR to restore clarity and tighten the low end on heavily processed or overdubbed material. The effect is subtle but adds coherence and perceived bass energy without any

level change. Works independently of both processors – try it with everything else turned off first to hear the difference.

Alternative bass character: Toggle BB Phase on to flip the sidechain polarity for a completely different bass enhancement character. The effect is dramatic – you may need to readjust Drive and Amount to taste. Try both settings and pick whichever suits the material better.

Consistent bass on varying material: Switch on BB AutoTrace when working with full mixes or live recordings where the bass level varies. The processor adapts automatically so the enhancement stays even across quiet and loud passages, without having to ride the Drive control.

Denser harmonics: Switch Exc Density to High for a hotter, more consistent harmonic character. Quiet signals get more excitement while peaks are controlled. Useful when you want the exciter to work harder across a wider dynamic range, or when high Harmonics settings cause clipping in Normal mode.

Support

For technical support, updates, and additional information:

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